

WEEK – 4

DAA assignment

Celebrity problem

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```
#include <stdio.h>
```

```
int findCelebrity(int n, int knows[n][n]) {
```

```
    int candidate = 0;
```

```
    // Step 1: Find a possible celebrity
```

```
    for (int i = 1; i < n; i++) {
```

```
        if (knows[candidate][i] == 1) {
```

```
            candidate = i;
```

```
        }
```

```
    }
```

```
    // Step 2: Verify the candidate
```

```
    for (int i = 0; i < n; i++) {
```

```
        if (i == candidate)
```

```
            continue;
```

```
    // Candidate should not know anyone
```

```
    // Everyone else should know candidate
```

```
    if (knows[candidate][i] == 1 ||
```

```
        knows[i][candidate] == 0) {
```

```
        return -1;
```

```
    }
```

```

    }

    return candidate;
}

int main() {
    int n = 4;

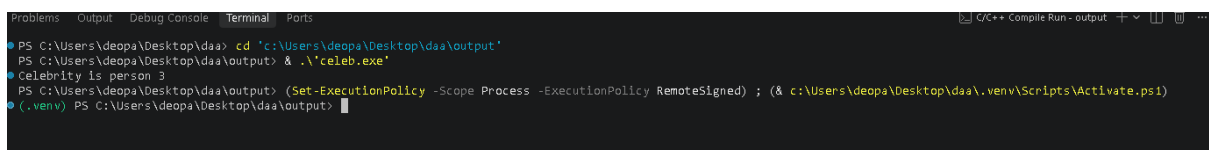
    int knows[4][4] = {
        {0, 1, 1, 1},
        {0, 0, 1, 1},
        {0, 0, 0, 1},
        {0, 0, 0, 0}
    };

    int celebrity = findCelebrity(n, knows);

    if (celebrity == -1)
        printf("No celebrity found\n");
    else
        printf("Celebrity is person %d\n", celebrity);

    return 0;
}

```



```

Problems Output Debug Console Terminal Ports
PS C:\Users\deopa\Desktop\daa> cd "c:\Users\deopa\Desktop\daa\output"
PS C:\Users\deopa\Desktop\daa\output> & .\celeb.exe
Celebrity is person 3
PS C:\Users\deopa\Desktop\daa\output> (Set-ExecutionPolicy -Scope Process -ExecutionPolicy RemoteSigned) ; (& c:\Users\deopa\Desktop\daa\venv\Scripts\Activate.ps1)
(.venv) PS C:\Users\deopa\Desktop\daa\output>

```